

BACKLOG SUBJECT EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	: B.E. - CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)	Semester	: III
Course Name	: Data Base Management Systems (Integrated)	Max. Marks	: 100
Course Code	: CY32 / CI32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Discuss the roles and responsibilities of the different end users involved in DBMS. CO1 (05)
 - Construct an Entity-Relationship Diagram considering the Data requirements given below. Specify the structural constraints clearly based on the requirements specified. CO1 (09)
Each musician that records at Notown has an SSN, a name, an address, and a phone number. Poorly paid musicians often share the same address, and no address has more than one phone.
 - Each instrument that is used in songs recorded at Notown has a name (e.g., guitar, synthesizer, flute) and a musical key (e.g., C, B-flat, E-flat).
 - Each album that is recorded on the Notown label has a title, a copyright date, a format (e.g., CD or MC), and an album identifier.
 - Each song recorded at Notown has a title and an author.
 - Each musician may play several instruments, and a given instrument may be played by several musicians. Each album has a number of songs on it, but no song may appear on more than one album.
 - Each song is performed by one or more musicians, and a musician may perform a number of songs. Each album has exactly one musician who acts as its producer. A musician may produce several albums, of course.
 - Describe the Degree of a relationship type and Recursive relationship with suitable examples. CO1 (06)
- Design an Entity-Relationship Diagram considering the Data requirements given below. CO1 (07)
 - The Zoo has many types of animals. Every type has a unique name. Every animal of the same type has a unique animal ID. Animals in two types may have the same animal ID. Animals also have age and gender.
 - Animals may have diseases. The beginning time and the duration of a disease need to be recorded. A disease has a unique name. A type keeper takes care of only one type of animals. Every type may have many type keepers.
 - A type keeper may or may not be familiar with diseases. But every disease must be handled by at least one type keeper. Type keeper have name, employee ID, ssn, address and phone number.
 - All animals are in cages. Some cage may be empty. Every cage has a cage ID, space and height. A cage keeper may take care of many cages. Every non-empty cage must have at least one cage keeper. Empty cages don't need any cage keepers. Cage keepers have name, employee ID, ssn, address and phone number.

- b) Describe the three-schema architecture and discuss the need of mappings between schema levels. Discuss how do different data definition languages support this architecture. CO1 (07)
- c) Discuss the characteristics of the DBMS Approach when comparing with the traditional file systems. CO1 (06)

UNIT – II

- 3. a) Describe the aggregate functions and grouping in relational algebra with syntax and examples. CO2 (07)
 - b) Consider the schema given below: CO2 (08)
Flights(flno:integer, from:string, to:string, distance:integer, departs:time, arrives:time)
Aircraft(aid: integer, aname: string, cruisingrange: integer)
Certified(eid: integer, aid: integer)
Employees(eid: integer, ename: string, salary: integer)
Give the relational algebra expression for the following:
 - i. Find the names of pilots certified for some Boeing aircraft.
 - ii. Identify the flights that can be piloted by every pilot whose salary is more than Rs. 100,000.
 - iii. Find the eids of employees who make the highest salary.
 - iv. Find the eids of employees who are certified for the largest number of aircraft.
 - c) Assume that you have two relations R & S where R is parent relation and S is child relation. R has primary key values where S references them. In a situation you are about to delete some tuples in relation R which causes inconsistency in relation S. As a designer what solution you might suggest to avoid inconsistencies in relation S? CO2 (05)
- 4. a) Describe the steps of an algorithm for ER to Relational Mapping with suitable example. CO2 (06)
 - b) Explain union compatibility of two relations. Describe any two set operations of relational algebra. CO2 (06)
 - c) With respect to the Relational data model, explain the concept of primary keys and Foreign keys and the constraints applicable to this. Illustrate your answer with examples. CO2 (08)

UNIT – III

- 5. a) Consider the following relations for a database that keeps track of business trips of salespersons in a sales office. CO3 (04)
SALESPERSON(SID, Name, Start_year, Dept_no)
TRIP(SID, From_city, To_city, Departure_date, Return_date, Trip_id)
EXPENSE(Trip_id, Account#, Amount)
Write SQL Queries for the following
 - i. List the Sid and the name of the salesman who made trips to 'San Francisco'.
 - ii. List the total trip expenses incurred by the salesman with name "RIO".

- b) Consider the schema given below: CO3 (10)

EMP(Ssn, fname, lname, bdate, address, sex, salary, superssn, dno)

DEPT(dnumber, dname, mgrssn, mgrstartdate)

PROJECT(pnumber, pname, ploc, dnum)

WORKS_ON(essn, pno, hours)

Write the SQL statements for the following queries:

- Select the names of all female employees whose first letter of first name is 'J' and fourth letter is 'H'.
 - For each project on which more than two employees work, retrieve the project number, the project name, and the number of employees who work on that project.
 - Retrieve the names of all employees whose supervisor's supervisor has '888445555' as his Ssn.
 - Update the Dumber of 'Research' Department from 10 to 15. Explain any constraint violations occur. If so, how to resolve that?
 - Show how to add a new column named "middlename" be added to EMP table.
- c) Discuss the basic clauses required to execute a SQL Query with syntax and example. CO3 (06)

6. a) Person (name, age, gender) CO3 (08)

Frequents (name, pizzeria)

Eats (name, qty)

Serves (pizzeria, pizzaname, price)

Write SQL Queries for the following statements

- Find the names of all females who eat both mushroom and pepperoni pizza.
 - Find all pizzerias that serve at least one pizza.
 - Find the number of females who eat pizza less than \$10 and age greater than 45.
 - Find the pizzeria serving the cheapest pepperoni pizza.
- b) Differentiate between GROUP BY and WHERE clause. CO3 (06)
- c) Discuss the correlated nested queries with an example. CO3 (06)

UNIT- IV

7. a) Differentiate between 3NF and BCNF. For the given relation Patient: CO4 (09)

i. Find the keys of the relation.

ii. Check whether the relation is in 3NF and BCNF. If not decompose the relation into 3NF and BCNF.

Patient (Patno, PatName, appNo, time, doctor)

FD's:

{Patno->PatName,

Patno->Time,doctor

Time ->appNo

appNo->Patno}

- b) Assume a relation R is defined over the attributes R(A,B,C,D,E,F) and the functional dependencies are CO4 (06)
- $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow F\}$ and $G = \{A \rightarrow CD, E \rightarrow AF\}$.
Check whether G and H are equivalent.
- c) Discuss the different types of anomalies in handling the relations with an example. CO4 (05)

8. a) Prove the following inference rules. CO4 (06)
- IR4 (decomposition, or projective, rule): $\{X \rightarrow YZ\} \models X \rightarrow Y$.
 - IR5 (union, or additive, rule): $\{X \rightarrow Y, X \rightarrow Z\} \models X \rightarrow YZ$.
 - IR6 (pseudotransitive rule): $\{X \rightarrow Y, WY \rightarrow Z\} \models WX \rightarrow Z$.
- b) Explain the 2NF with an example. CO4 (05)
- c) A set of FD's for the relation $R\{A,B,C,D,E,F\}$ is $AB \rightarrow C, C \rightarrow A, BC \rightarrow D, ACD \rightarrow B, BE \rightarrow C, EC \rightarrow FA, CF \rightarrow BD, D \rightarrow E$. Find a minimum cover for this set of FD'S. Decompose the relation Using Relational Synthesis algorithm into 3NF. CO4 (09)

UNIT - V

9. a) With a neat diagram, write and explain the state transition diagram illustrating the states for transaction execution. CO5 (07)
- b) For each of the following schedules, state whether they are serializable and whether they are conflict-serializable. For each schedule, draw the corresponding precedence graph. If the schedule is conflict-serializable, show all the conflict-equivalent serial schedules. CO5 (06)
- i. $S1 = r1(A) \ r2(A) \ w1(A) \ w2(A)$
 - ii. $S2 = r1(A) \ r2(B) \ w3(A) \ r2(A) \ r1(B)$
 - iii. $S3 = r1(A) \ w2(A) \ w1(A) \ r3(A)$
- c) Describe the procedure of two phase locking? Will two phase locking result in serializable schedule? Will two phase locking result in deadlock? Demonstrate with an example. CO5 (07)
10. a) Which of the following schedules is (conflict) serializable? For each serializable schedule, determine the equivalent serial schedules. CO5 (08)
- i. $r1(X); r3(X); w1(X); r2(X); w3(X);$
 - ii. $r1(X); r3(X); w3(X); w1(X); r2(X);$
 - iii. $r3(X); r2(X); w3(X); r1(X); w1(X);$
 - iv. $r3(X); r2(X); r1(X); w3(X); w1(X);$
- b) What is the system log used for? What are the typical kinds of records in a system log? What are transaction commit points, and why are they important? CO5 (07)
- c) Discuss on The Incorrect Summary Problem occurs when the transaction is executed concurrently. CO5 (05)
